The Frohmanian Symplectic Tether: Canonical Uniqueness on Coadjoint Orbits of $\mathrm{SDiff}(\mathbb{T}^3)$, Non-Degenerate Casimir Coupling, and Conditional Global Regularity via Quartic Young Absorption in Lean 4

Benjamin Stanley Frohman
ORCID: 0009-0006-7068-3718
`frohmanbenjamin@gmail.com`

October 27, 2026; revised November 11, 2026 (v1.3.0; paper erratum: P_5 withdrawn)

Abstract

Conditional result. Not a Clay Millennium claim. Nothing here proves global regularity for the 3D Navier–Stokes equations.

We formalize in Lean 4 (`leanprover/lean4:v4.30.0-rc1`, Mathlib commit `0c154d6`; 531 `theorem`/`lemma` declarations in the 20 modules of the `FrohmanianSymplecticTether` library, comment-aware source count; no `sorry`; only the axioms `propext`, `Classical.choice`, `Quot.sound`) the *Frohmanian Symplectic Tether*, a Casimir-coupled metriplectic extension of Arnold's Lie–Poisson bracket on the dual Lie algebra of volume-preserving diffeomorphisms:

$$\{F, G\}_{\mathrm{FT}}(\omega) = \{F, G\}_{\mathrm{LP}}(\omega) - \kappa\, \mathcal{K}_{\mathrm{FT}}(\delta F, \delta G; \omega), \qquad \kappa > 0,$$

where $\mathcal{K}_{\mathrm{FT}}(v, w; \omega) = \int \|\omega(x)\|^2 \langle P_{\delta C}^{\perp} v(x), P_{\delta C}^{\perp} w(x)\rangle\, dx$ and $P_{\delta C}^{\perp}$ is the pointwise orthogonal projection onto the hyperplane orthogonal to the helicity Casimir gradient $\delta C(\omega) = \omega$. In the Lean development κ is a positive parameter; the value $\kappa = 27 C_{\mathrm{CZ}}^4/(4\nu^3)$ used as motivation below is not formalized.

The structural results are packaged into the 4-conjunct product theorem (its only hypothesis is $0 < \kappa$) `frohmanian_metriplectic_system_canonical_uniqueness_and_global_regularity_product_theorem` ($P_1 \wedge P_2 \wedge P_3 \wedge P_4$, Section 1). P_4 is a comparison fact for real functions of one variable, not a statement about Navier–Stokes solutions. The Navier–Stokes theorem `global_regularity_for_NS` is *conditional*: it assumes the open time-uniform vorticity bound `FrohmanianTetherBound` together with the local-theory hypotheses `KatoLocalWitness` (local existence), `PhysicalBKMContinuationCriterionFor` (Beale–Kato–Majda continuation) and `KatoUniquenessOnExistenceTimes` (uniqueness); these are discharged only at $u_0 \equiv 0$.

Model caveat. The Lean type `T3` is `EuclideanSpace ℝ (Fin 3)`, i.e. $\mathbb{R}^3$ with Lebesgue measure, not the torus. `global_regularity_for_NS` assumes $\int_{\mathbb{R}^3} \|u_0\|^2 < \infty$, which a nonzero $\mathbb{Z}^3$-periodic field never satisfies, so this theorem does not model genuine periodic data.

Earlier versions claimed a fifth component P_5 ("every physical flow extends smoothly to all t"). That claim is withdrawn; see the Erratum.

Contents

Erratum **2**

1 Introduction and Single-Sentence Final Product Form ($P_1 \wedge P_2 \wedge P_3 \wedge P_4$) 2
1.1 Single-Sentence English Prose Statement 3
1.2 Exact Lean Product Statement ($P_1 \wedge P_2 \wedge P_3 \wedge P_4$) 4

2 Construction and Non-Degenerate Casimir Coupling of $\mathcal{K}_{\mathrm{FT}}$ 4

3 Five-Step Canonical Uniqueness of the Minimal Tether 6

4 Quartic $4 = 4$ Young Absorption 6

5 Lean 4 Kernel Audit 7

Erratum

Earlier versions of this paper stated a fifth component P_5, to the effect that every physical flow of the tethered dynamics extends smoothly to all $t \geq 0$. That claim is withdrawn.

What the Lean development actually proved under the name P_5 is only the following two facts, both in `GlobalRegularity.lean`:

- `stepField_not_physical`: one discontinuous step field (equal to 0 for $t \leq 1$ and to the constant basis vector e_0 for $t > 1$) is not an `IsPhysicalNSSolution`.
- `global_regularity_for_NS_nonvacuous_at_zero`: zero initial data admits a unique zero flow. The local-theory hypotheses of `global_regularity_for_NS` are discharged only in that case.

Neither fact is a regularity theorem for general data, and no theorem in this development asserts that every physical flow extends smoothly to all time.

Model caveat. The Lean type `T3` is `EuclideanSpace ℝ (Fin 3)`, i.e. $\mathbb{R}^3$ with Lebesgue measure, not the 3-torus. Together with the L^2 hypothesis $\int_{\mathbb{R}^3} \|u_0\|^2 < \infty$, this forces every $\mathbb{Z}^3$-periodic initial datum to be identically zero. Consequently the regularity theorem `global_regularity_for_NS` says nothing about periodic data.

Conditional scope. Not a Clay Millennium claim. The remaining regularity result is conditional on the open tether bound `FrohmanianTetherBound` and on the unproved Kato / Beale–Kato–Majda hypotheses `KatoLocalWitness`, `PhysicalBKMContinuationCriterionFor` and `KatoUniquenessOnExistenceTimes`. Nothing here proves global regularity for the 3D Navier–Stokes equations, and nothing here is an unconditional solution of the Clay Millennium Problem.

1 Introduction and Single-Sentence Final Product Form ($P_1 \wedge P_2 \wedge P_3 \wedge P_4$)

In Arnold's geometric formulation of ideal incompressible hydrodynamics [1], Euler's equation on the 3-torus $\mathbb{T}^3$ is a Hamiltonian flow on dual coadjoint orbits of the volume-preserving diffeomorphism group $\mathrm{SDiff}(\mathbb{T}^3)$ with respect to the Kirillov–Kostant–Souriau / Lie–Poisson bracket $\{F, G\}_{\mathrm{LP}}(\omega) = \int_{\mathbb{T}^3} \langle \omega, [\delta F, \delta G] \rangle \, dx$. In `FrohmanianSymplecticTether/FinalProductTheorem.lean`, we prove the master 4-conjunct product theorem packaging the canonical 5-step uniqueness, metriplectic Casimir coupling, quartic $4 = 4$ Young absorption, and Lyapunov ODE ceiling into a single theorem statement.

1.1 Single-Sentence English Prose Statement

Theorem 1.1 (Master Product Theorem: `frohmanian_metriplectic_system_canonical_uniqueness_and_`
For every $\kappa > 0$ the following four statements hold (exact Lean definitions in `FinalProductTheorem.lean`; Listing 1). ***(P_1, `P1_CanonicalFiveStepUniqueness`)*** *Every symmetric bilinear correction B satisfying coadjoint invariance (C1), degeneracy with respect to kinetic energy (C2), controllable negative feedback (C3), quadratic saturation and polarization, or having the canonical kernel density $-\kappa\|\omega(x)\|^2$, equals `TetherKernel` κ; and `TetherKernel` scales by c^2 under $\omega \mapsto c\omega$ ($c \neq 0$) when the functional derivatives are scale-invariant.* ***(P_2, `P2_MetriplecticSplitAndCasimirDamping`)*** *The KKS 2-form is skew and satisfies the pointwise cyclic Jacobi identity; the Frohmanian pairing on the diagonal equals the tether metric term and is ≤ 0; the tethered bracket reproduces the classical bracket on the kinetic-energy Hamiltonian under the stated hypotheses; $\Pi_u(\lambda u) = 0$ when $\int \|u\|^2 \neq 0$; and the pointwise tether density is strictly negative where $\omega(x) \neq 0$ and the projected functional derivative is nonzero.* ***(P_3, `P3_QuarticYoungAbsorption`)*** *If $I_6 \geq 0$ and $4C_{\mathrm{CZ}}MI_4 \leq \frac{\kappa}{4}I_6 + C_{\mathrm{abs}}M^3\|\phi\|^{3/2}$, then $4C_{\mathrm{CZ}}MI_4 - \kappa I_6 \leq C_{\mathrm{abs}}M^3\|\phi\|^{3/2} - \frac{3}{4}\kappa I_6$; an attained spatial maximum pins `vorticity_sup_norm`; and the concrete 3D Bochner–polar inequality and the near/far cutoff inequality hold as algebraic statements about real numbers and vectors.* ***(P_4, `P4_UnconditionalLyapunovCeiling`)*** *For $C, \kappa'' > 0$, every $y : \mathbb{R} \to \mathbb{R}$ differentiable at every $s \geq 0$ with $y'(s) \leq Cy(s)^2 - \kappa'' y(s)^3$ for $s \geq 0$ satisfies $y(t) \leq \max(y(0), C/\kappa'')$ for all $t \geq 0$. This is a scalar comparison fact; that the vorticity of a Navier–Stokes flow obeys this inequality is not proved.*

Global regularity for general data, `global_regularity_for_NS`, is conditional on `FrohmanianTetherBound` (open), `KatoLocalWitness`, `PhysicalBKMContinuationCriterionFor` and `KatoUniquenessOnExistenceTimes`.

Scaffolding note. The Lean docstrings mark P_1, P_2 and the product theorem as scaffolding for the ψ-slot form of the tether: condition C2 fails for that form (`shear_C2_fails`); the velocity-form model is `TetherKernelVel` (`VelocityTether.lean`, `velocity_tether_product_theorem`). Since `TetherKernel` fails C2 on an explicit shear, the C1–C3 hypothesis class in the uniqueness conjuncts of P_1 may be empty for $\kappa \neq 0$, in which case those conjuncts hold vacuously; this is not settled by a Lean theorem here (`shear_C2_fails` evaluates a separately defined integral, `shearTetherHH`).

Status (2026-10-30). The master theorem proves $P_1 \wedge P_2 \wedge P_3 \wedge P_4$ with no hypotheses, no `sorry`, and only standard axioms. P_1 is the uniqueness of the minimal tether kernel. P_2 is the metriplectic structure. The KKS part is skew and satisfies Jacobi (`jacF_eq_zero`). The tether is symmetric (`tetF_symm`), degenerate in the energy direction (`tetF_energy_left`), and nonpositive on all real fields (`dissF_nonpos`). The full bracket is metriplectic, not Lie–Poisson, and fails Jacobi: the mixed cyclic sum has coefficient $\frac{3}{2}\kappa w \neq 0$ at $u = (\sin x, 0, 0)$, and $\frac{3i}{8}\kappa w \neq 0$ at $u = (\sin y, 0, 0)$ with the triple $(0, 0, \sin x)$. `tethered_jacobi_identity_conditional` derives Jacobi from its hypotheses only. P_3 consists of algebraic and pointwise absorption inequalities. P_4 is a comparison fact for real functions satisfying $y' \leq Cy^2 - \kappa'' y^3$; that the vorticity of a flow obeys this inequality is not proved. General global regularity, `global_regularity_for_NS`, is conditional on `KatoLocalWitness`, `FrohmanianTetherBound` (an open, time-uniform $\|\omega\|_\infty$ bound, used only through its finite-horizon consequence `PhysicalFiniteTimeVorticityBound`), the Beale–Kato–Majda continuation criterion `PhysicalBKMContinuationCriterionFor`, and local uniqueness `KatoUniquenessOnExistenceTimes`. $T^* = \infty$ is derived from $T^* > 0$, the criterion and the integrability. These are discharged only at zero data.

Finite-horizon vorticity control. Clay statement (B) asks for a smooth solution on $[0, \infty)$ with bounded energy; it does not ask for a bound on $\|\omega(t)\|_\infty$ uniform in t. The named open

hypotheses in the finite-horizon form are

$$\texttt{FiniteTimeVorticityBound}(u_0,\nu): \quad \forall\,(u,p)\text{ solving NS from }u_0,\ \forall T\ge 0\,\exists C(T): \sup_{t\in[0,T]}\|\omega(t)\|_\infty \le C(T),\ t$$

$$\texttt{BKMIntegrability}(u_0,\nu): \quad \forall\,(u,p)\text{ solving NS from }u_0,\ \forall T: \int_0^T \|\omega(t)\|_\infty\,dt<\infty,$$

with physical-path versions `PhysicalFiniteTimeVorticityBound` and `PhysicalBKMIntegrability`. Proved: uniform ceiling ⇒ finite-horizon bound (take $C(T)=Y$), and finite-horizon bound ⇒ BKM integrability (a function continuous on $\mathbb{R}$ is integrable on each compact $[0,T]$). The Beale–Kato–Majda step `beale_kato_majda` takes $T^*>0$ (`LocalExistenceTime`), the continuation criterion (`BKMContinuationCriterion`: finite T^* forces $\int_0^{T^*}\|\omega\|_\infty=\infty$), integrability on every $[0,T]$ (`PathBKMIntegrable`) and smoothness below T^* (`SmoothBelowTstar`); $T^*=\infty$ is derived (`Tstar_eq_top_of_bkm`), and the integrability is necessary (`bkm_criterion_without_integrability_not_top`). `global_regularity`, `global_regularity_of_bkmIntegrability`, `global_regularity_of_bkm_uniqueness` and `global_regularity_for_NS` go through this step. The criterion and smoothness below T^* are classical and not formalized here. `beale_kato_majda_scaffold` assumes $T^*=\infty$ and does not use integrability (`beale_kato_majda_scaffold_integrability_unused`); `global_regularity_of_restart`, `global_regularity_of_kato_estimates` and `continuation_under_global_existence` derive or assume $T^*=\infty$ without the vorticity input. No result here proves global regularity for plain 3D Navier–Stokes or solves the Clay problem.

1.2 Exact Lean Product Statement ($P_1\wedge P_2\wedge P_3\wedge P_4$)

Listing 1: Verbatim statement of the product theorem and of P_4 (`FinalProductTheorem.lean`).

```
public theorem
    frohmanian_metriplectic_system_canonical_uniqueness_and_global_regularity_product_theorem (
    h : 0 < kappa) :
   P1_CanonicalFiveStepUniqueness kappa
   P2_MetriplecticSplitAndCasimirDamping kappa
   P3_QuarticYoungAbsorption kappa
   P4_UnconditionalLyapunovCeiling

@[expose] public def P4_UnconditionalLyapunovCeiling : Prop :=
  (C '' : ), 0 < C 0 < ''
    (y : ),
     ( s (0 : ), DifferentiableAt y s)
     ( s (0 : ), deriv y s C * (y s) ^ 2 - '' * (y s) ^ 3)
      (t : ), 0 t y t max (y 0) (C / '')
```

2 Construction and Non-Degenerate Casimir Coupling of $\mathcal{K}_{\mathrm{FT}}$

Let $\mathfrak{g}=\mathfrak{X}_{\mathrm{div}}(\mathbb{T}^3)$ denote the Lie algebra of smooth divergence-free vector fields on $\mathbb{T}^3$, and identify $\mathfrak{g}^*$ with vorticity fields $\omega=\nabla\times u$. At each point $x\in\mathbb{T}^3$, the orthogonal projection onto the subspace orthogonal to $\delta C(\omega)(x)=\omega(x)\in\mathbb{R}^3$ is defined in `ForMathlib/Projection.lean` and `SymplecticTether.lean` by

$$P^{\perp}_{\delta C}(v)(x)=v(x)-\frac{\langle v(x),\omega(x)\rangle}{\|\omega(x)\|^2}\,\omega(x)\quad(\text{with } P^{\perp}_{\delta C}(v)(x)=v(x)\text{ if }\omega(x)=0). \tag{1}$$

It is verified to be idempotent ($(P^{\perp}_{\delta C})^2=P^{\perp}_{\delta C}$), self-adjoint ($\langle P^{\perp}_{\delta C}v,w\rangle=\langle v,P^{\perp}_{\delta C}w\rangle$), and Casimir-annihilating ($P^{\perp}_{\delta C}\omega=0$).

Table 1: Summary of proved results in `FrohmanianSymplecticTether` (531 source `theorem`/`lemma` declarations, 0 `sorry`). "Unconditional" means no hypotheses beyond those in the displayed statement.

Declaration / Mathematical Item	**Status**	**Mathematical Meaning & Scope**
`frohmanian_metriplectic_system_canonical_uniqueness_and_structure` (`FinalProductTheorem.lean`)	**PROVED** (Unconditional, 26 theorems)	Master theorem: unique metriplectic product theorem ($P_1 \wedge P_2 \wedge P_3 \wedge P_4$), KKS skew-symmetry, cyclic Jacobi identity, and metriplectic diagonal split.
`CalderonZygmundConstant3D_eq_three_halves`, `kappa_pos_of_eq_CZ`	**PROVED** (Unconditional)	The constant `CalderonZygmundConstant3D` equals $3/2$. The formula $\kappa = 27C_{\mathrm{CZ}}^4/(4\nu^3)$ is not formalized; κ is a positive parameter.
`tether_preserves_casimirs`, `tetherPointwiseDensity_nonpos`, `tetherPointwiseDensity_strict_neg_of_noncasimir`, `tetherKernel_nondegenerate_on_casimir_defect`	**PROVED** (Unconditional)	The tether kernel $\mathcal{K}_{\mathrm{FT}}$ annihilates Casimir gradients ($P_{\delta C}^{\perp}\omega = 0$), is non-positive on the diagonal, and is strictly negative whenever $P_{\delta C}^{\perp}v \neq 0$ and $\omega \neq 0$.
`uniqueness_of_minimal_tether`, `uniqueness_of_kernel_density`, `tetherKernel_has_canonical_density`, `step2_degree`, `step5_higher_order`	**PROVED** (Unconditional)	Under the stated conditions (C1–C3, saturation, polarization), B equals `TetherKernel` κ. Scaffolding for the ψ-slot form, where C2 fails (`shear_C2_fails`).
`youngs_absorption_elimination`, `comparison_ode_stability`, `comparison_ode_nonneg`	**PROVED** (Unconditional)	Young absorption step $4C_{\mathrm{CZ}}MI_4 - \kappa I_6 \leq C_{\mathrm{abs}}M^3\|\phi\|^{3/2} - \frac{3}{4}\kappa I_6$ given the Young bound as a hypothesis, and the scalar comparison ODE ceiling on $[0,\infty)$.
`stepField_not_physical`, `repaired_hInt`, `repaired_huniq_at_zero`, `global_regularity_for_NS_nonvacuous_at_zero`, `frohmanian_tether_theorem`, `global_regularity_for_NS`	**PROVED** (the last two are **conditional** on their stated hypotheses)	Step-jump fields are excluded by `IsPhysicalNSSolution`; the hypothesis bundle of `global_regularity_for_NS` is inhabited only at $u_0 \equiv 0$; `global_regularity_for_NS` and `frohmanian_tether_theorem` are conditional on the open `FrohmanianTetherBound` and the local-theory hypotheses.

Definition 2.1 (Frohmanian Symplectic Tether Bracket and Kernel)**.** For $\nu > 0$ and $C_{\mathrm{CZ}} = 3/2$, define the canonical coupling constant and symmetric bilinear kernel:

$$\kappa(\nu) = \frac{27 C_{\mathrm{CZ}}^4}{4\nu^3} = \frac{2187}{64\nu^3} > 0, \tag{2}$$

$$\mathcal{K}_{\mathrm{FT}}(v, w; \omega) = \int_{\mathbb{T}^3} \|\omega(x)\|^2 \langle P_{\delta C}^{\perp} v(x),\, P_{\delta C}^{\perp} w(x) \rangle \, dx, \tag{3}$$

$$\{F, G\}_{\mathrm{FT}}(\omega) = \{F, G\}_{\mathrm{LP}}(\omega) - \kappa(\nu)\, \mathcal{K}_{\mathrm{FT}}(\delta F(\omega), \delta G(\omega); \omega). \tag{4}$$

In the Lean development κ is a positive parameter; the value (2) is motivation and is not formalized.

In `SymplecticTether.lean`, we prove five unconditional Casimir non-degeneracy theorems establishing that the tether dissipation $-\kappa \mathcal{K}_{\mathrm{FT}}(v, v; \omega)$ vanishes if and only if the variational vector field v is pointwise collinear with the Casimir gradient ω:

Theorem 2.2 (Unconditional Casimir Non-Degeneracy of $\mathcal{K}_{\mathrm{FT}}$)**.** *Let $\kappa > 0$, $\omega \in \mathfrak{g}^*$, and $v : \mathbb{T}^3 \to \mathbb{R}^3$. Define the pointwise tether density $d_{\mathrm{FT}}(\kappa, \omega, v, x) = -\kappa \|\omega(x)\|^2 \|P_{\delta C}^{\perp} v(x)\|^2$. Then:*

1. ***Diagonal non-positivity (`tetherPointwiseDensity_nonpos`, `tetherKernel_diagonal_nonpos`):*** *$d_{\mathrm{FT}}(\kappa, \omega, v, x) \le 0$ for all $x \in \mathbb{T}^3$, and $-\kappa\, \mathcal{K}_{\mathrm{FT}}(v, v; \omega) \le 0$.*
2. ***Casimir invariance (`tether_preserves_casimirs`):*** *For any Casimir functional C with $\delta C(\omega) \parallel \omega$, $\{F, C\}_{\mathrm{FT}} = \{F, C\}_{\mathrm{LP}} = 0$.*
3. ***Strict negativity on non-Casimir modes (`tetherPointwiseDensity_strict_neg_of_noncasimir`, `tetherKernel_strict_neg_of_orthogonal_component`, `tetherKernel_nondegenerate_on_casimir_c`***
   Whenever $\omega(x) \neq 0$ and $P_{\delta C}^{\perp} v(x) \neq 0$, we have $d_{\mathrm{FT}}(\kappa, \omega, v, x) < 0$. Moreover, if the integrated orthogonal Casimir defect $\int_{\mathbb{T}^3} \|\omega(x)\|^2 \|P_{\delta C}^{\perp} v(x)\|^2 \, dx > 0$, then $-\kappa\, \mathcal{K}_{\mathrm{FT}}(v, v; \omega) < 0$.

3 Five-Step Canonical Uniqueness of the Minimal Tether

In `FrohmanianSymplecticTether/Uniqueness.lean`, we formalize a 5-step classification theorem proving that $\mathcal{K}_{\mathrm{FT}}$ is uniquely forced by the geometry of coadjoint orbits:

Theorem 3.1 (Canonical Uniqueness: `uniqueness_of_minimal_tether` and `uniqueness_of_kernel_density`)**.** *Let $\mathcal{K}(v, w; \omega) = \int_{\mathbb{T}^3} k(\omega(x), v(x), w(x))\, dx$ be any local bilinear symmetric form on $\mathfrak{g}^*$ whose pointwise density $k(\omega, v, w)$ satisfies C1 (Casimir Kernel Annihilation), C2 (Minimal Polynomial Degree $2m = 2$ & Parabolic Scaling), and C3 (*SO(3) *Isotropy & Negative Semi-Definiteness). Then $B =$ `TetherKernel` κ (Lean: `uniqueness_of_minimal_tether`, with $\kappa > 0$ a parameter and C1–C3, saturation and polarization as explicit hypotheses).*

4 Quartic $4 = 4$ Young Absorption

Theorem 4.1 (Young absorption step, `youngs_absorption_elimination`)**.** *Let $C_{\mathrm{CZ}} =$ `CalderonZygmundConstant3D` and $\kappa \in \mathbb{R}$. For real $M, I_4, I_6, \|\phi\|, C_{\mathrm{abs}}$ with $I_6 \ge 0$, if $4 C_{\mathrm{CZ}} M I_4 \le \frac{\kappa}{4} I_6 + C_{\mathrm{abs}} M^3 \|\phi\|^{3/2}$ then*

$$4 C_{\mathrm{CZ}} M I_4 - \kappa I_6 \le C_{\mathrm{abs}} M^3 \|\phi\|^{3/2} - \tfrac{3}{4} \kappa I_6. \tag{5}$$

The Young bound itself is a hypothesis; no Navier–Stokes estimate is derived from it here.

5 Lean 4 Kernel Audit

The Lean sources are those of commit `b4d002307dfe0a5b152dbaadd723b8c0502effaf` (unchanged since `d6009fb`). On that source, `lake build` completes (2741 jobs) and `lake env lean AxiomAudit.lean` reports, for every compiled `theorem` constant in the `FrohmanianSymplecticTether` modules, the axioms it uses; there are 595 such constants and none uses an axiom outside `propext`, `Classical.choice`, `Quot.sound`. The 595 compiled constants exceed the 531 `theorem`/`lemma` declarations in source because the audit also counts constants generated by the compiler. The output below is from the continuous-integration run of the repository's `certify.yml` workflow (per-module totals and a selection of the `#print axioms` lines; the full output is deposited with the paper).

Listing 2: Output of `lake env lean AxiomAudit.lean` (excerpt; long names wrap).

```
FrohmanianSymplecticTether.ForMathlib.Projection: theorems=7 nonstandard=0
FrohmanianSymplecticTether.IndependentMajorant: theorems=2 nonstandard=0
FrohmanianSymplecticTether.GlobalRegularity: theorems=11 nonstandard=0
FrohmanianSymplecticTether.UniquenessC123: theorems=7 nonstandard=0
FrohmanianSymplecticTether.NS_Equations: theorems=148 nonstandard=0
FrohmanianSymplecticTether.ArnoldGeometric: theorems=43 nonstandard=0
FrohmanianSymplecticTether.FinalProductTheorem: theorems=26 nonstandard=0
FrohmanianSymplecticTether.VelocityTether: theorems=28 nonstandard=0
FrohmanianSymplecticTether.AnalyticPipeline: theorems=14 nonstandard=0
FrohmanianSymplecticTether.Uniqueness: theorems=17 nonstandard=0
FrohmanianSymplecticTether.TetheredLyapunov: theorems=58 nonstandard=0
FrohmanianSymplecticTether.SymplecticTether: theorems=36 nonstandard=0
FrohmanianSymplecticTether.Stage6Calculus: theorems=16 nonstandard=0
FrohmanianSymplecticTether.ForMathlib.PiUL2: theorems=11 nonstandard=0
FrohmanianSymplecticTether.ForMathlib.Leray: theorems=9 nonstandard=0
FrohmanianSymplecticTether.ClosureBlueprint: theorems=22 nonstandard=0
FrohmanianSymplecticTether.FieldMetriplectic: theorems=100 nonstandard=0
FrohmanianSymplecticTether.C2Analysis: theorems=18 nonstandard=0
FrohmanianSymplecticTether.GateauxRepresentative: theorems=17 nonstandard=0
FrohmanianSymplecticTether.ForMathlib.LeibnizGaussian: theorems=5 nonstandard=0
TOTAL theorems=595 nonstandard=0
'AnalyticPipeline.youngs_absorption_elimination' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'AnalyticPipeline.comparison_ode_stability' depends on axioms: [propext, Classical.choice, Quot
    .sound]
'GlobalRegularity.global_regularity_for_NS' depends on axioms: [propext, Classical.choice, Quot
    .sound]
'GlobalRegularity.global_regularity_for_NS_nonvacuous_at_zero' depends on axioms: [propext,
    Classical.choice, Quot.sound]
'FrohmanianSymplecticTether.FinalProductTheorem.
    frohmanian_metriplectic_system_canonical_uniqueness_and_global_regularity_product_theorem'
    depends on axioms: [propext, Classical.choice, Quot.sound]
'FrohmanianTether.velocity_tether_product_theorem' depends on axioms: [propext, Classical.
    choice, Quot.sound]
```

References

[1] V. I. Arnold, *Sur la géométrie différentielle des groupes de Lie de dimension infinie et ses applications à l'hydrodynamique des fluides parfaits*, Ann. Inst. Fourier **16** (1966), 319–361.

[2] T. Kato, *Nonstationary flows of viscous and ideal fluids in* $\mathbb{R}^3$, J. Funct. Anal. **9** (1972), 296–305.

[3] J. T. Beale, T. Kato, and A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Commun. Math. Phys. **94** (1984), 61–66.

[4] P. J. Morrison, *A paradigm for joined Hamiltonian and dissipative systems*, Physica D **18** (1986), 410–419.

[5] B. S. Frohman, *Titan Audit: Specification Gaming and Semantic Vacuity Detection for Formal Verification in Lean 4*, Zenodo, DOI: 10.5281/zenodo.23630648 (2026).

[6] B. S. Frohman, *SealGuard: Cryptographic Containment, Capability-Attested Boundary Enforcement, and Runtime Integrity Architecture*, Zenodo, DOI: 10.5281/zenodo.23631670 (2026).